

- 10** Steam reforming of methane is the main source of production of hydrogen in the industry. There are two steps in the process, both being in equilibrium in closed systems:



Which statement about **Step 1** and **Step 2** is correct?

- A** Increasing the pressure increases the equilibrium constant for **Step 2**.
- B** Increasing the temperature increases the equilibrium constant for **Step 1**.
- C** Addition of a catalyst increases both the yield and rate of production of hydrogen gas.
- D** Increasing the temperature decreases the rate constant for the forward reaction in **Step 2**.